

Sharp local stability and critical-exponent sensitivity of Ptolemaic split metrics

All-cardinality first-order constants and the four-point transition

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Assurance. The new local theorems below are proved from explicit matrix calculations and a finite collection of rational-function certificates. They do not require the companion all-size theorem. Calling the listed split points the complete equality family for $m \geq 6$ uses the supplied v0.2 classification and its stated dependencies. The new four-point classification imports the four-point threshold, strict descent and Ercan's scalar estimate and coefficient transport. No proof-assistant verification, authenticated independent specialist review or exhaustive priority search is claimed. The complete v0.2 archive is preserved unchanged.

Abstract

We determine the sharp first-order stability constants at every binding complete split metric of cardinality $m \geq 6$, and also at the five-point split. Under mean-cross-distance normalisation, the negative-type gap is comparable to the maximum distance perturbation with exact lower and upper one-sided asymptotic constants. Eighteen rational-function dual certificates prove the result in every larger dimension; thirty-six finite certificates cover the small exceptions. Their nonnegativity for the entire parameter range is certified by polynomial coefficient expansions, not inferred from tests of finitely many dimensions. We give explicit quadratic remainder bounds and prove that every direction in the linearised active-constraint cone is realised by a Ptolemaic curve. The resulting coefficient has three stable parity-dependent formulas after cardinality eight. We then calculate the first variation of the metric's own supremal negative-type exponent, obtaining sharp directional sensitivity constants. The three- and four-point equality families are also identified; at the four-point star, unequal-arm perturbations produce a quadratic gap, so the linear stability phenomenon does not extend to that cardinality. Finally, a quotient argument identifies the full kernel at collision limits and an explicit subcritical gap bound quantifies strictness for separated smaller-cardinality metrics. No numerical value for the global separated-class constant is asserted.

1. Scope, notation and imported results

A symmetric array d on distinct labelled points is a Ptolemaic metric if its distances are positive off the diagonal, satisfy the triangle inequalities, and satisfy

$$d_{ij}d_{kl} \leq d_{ik}d_{jl} + d_{il}d_{jk}.$$

The metric $s = \text{CS}(a, r)$ has parts A, B of sizes a, r : its distances are one within A and across the parts, and two within B . Put $m = a + r$ and

$$T = \frac{r(a+1)}{a(r-1)}, \quad p = \log_2 T.$$

The binding pairs considered here are $(a, r) = (k, k+1)$, (k, k) and $(k, k+2)$, with $m \geq 6$, together with $(2, 3)$. Thus $a \geq 2$, $r \geq 3$, and $0 < p < 2$. For the first three families p is exactly the universal threshold p_m in the companion paper; in the five-point case this is Ercan's endpoint. The supplied v0.2 manuscript classifies all equality metrics for $m \geq 6$ as similarities of these splits. We do not re-prove that classification.

Let $\Pi = I - \mathbf{1}\mathbf{1}^\top/m$ and define

$$\Delta_p(d) = \min_{z \perp \mathbf{1}, \|z\|_2=1} -z^\top d^p z.$$

Powers are entrywise. This is a restricted eigenvalue; the automatic centring zero on the full space is not included. At the split, the restriction of $-s^p$ has eigenvalues 0, 1 with multiplicity $a-1$, and T with

multiplicity $r - 1$. The unit null direction is proportional to $(r\mathbf{1}_A, -a\mathbf{1}_B)$, with squared normalising factor arm .

Normalise the mean of the ar cross-distances to one. Write $e = d - s$, with coordinates y_{ij} within A , x_{ib} across, and t_{bc} within B , and set $h = \|e\|_\infty$. The normalisation is $\sum x_{ib} = 0$. Throughout, a local statement is at a fixed labelled split; constants may depend on (a, r) .

Attribution. The transport of a zero-sum coefficient vector through metric inversion is Ercan's Theorem 8.1, and the sign-changing choice of vertex is his Theorem 8.2 [E]. The equality-potential observation in v0.2 is used below, with these attributions retained. The threshold and Lorentzian correspondence originate in [B]; strict negative-type descent is [LW]. Matrix perturbation is standard [K]. The new work in this supplement concerns the complete tangent-cone calculation, the all-size sharp constants and the critical-exponent expansion. The short small-cardinality consequences are not presented as independently verified historical novelties.

2. Main local theorem

Define the sum of all linearised active slacks by

$$\ell(e) = -r(r-1) \sum_{i < j \in A} y_{ij} - \frac{a(a+1)}{2} \sum_{b < c \in B} t_{bc}.$$

Set

$$\gamma = \frac{2p}{ma(r-1)}, \quad U_{a,r} = \frac{pr(3a-1)}{2m}, \quad A_{a,r} = \gamma c_{a,r},$$

where the rational constant $c_{a,r}$ is

Pair or family	Range	$c_{a,r}$
(2, 3)	five-point local result	6/5
(2, 4)	six points	3/2
(3, 3)	six points	9/5
(3, 4)	seven points	3
(3, 5)	eight points	60/11
(4, 4)	eight points	5
(k, k)	$k \geq 5$	$k^2/2$
($k, k+1$)	$k \geq 4$	$k(k+1)/2$
($k, k+2$)	$k \geq 4$	$k(k+1)/2$

Theorem 2.1 (sharp first-order stability in all these dimensions). Put

$$b_{a,r} = p(m-1) + 18p^2(m-1)^2 + \frac{3pr(a-1)}{m}, \quad \rho_{a,r} = \min \left\{ \frac{1}{8}, \frac{1}{12p(m-1)} \right\}.$$

Every Ptolemaic d in the preceding normalisation with $h \leq \rho_{a,r}$ satisfies

$$A_{a,r}h - b_{a,r}h^2 \leq \Delta_p(d) \leq U_{a,r}h + p(m-1)h^2.$$

Both linear coefficients are sharp:

$$\liminf_{\substack{d \rightarrow s, d \neq s \\ d \text{ Ptolemaic}}} \frac{\Delta_p(d)}{h} = A_{a,r}, \quad \limsup_{\substack{d \rightarrow s, d \neq s \\ d \text{ Ptolemaic}}} \frac{\Delta_p(d)}{h} = U_{a,r}.$$

In particular $\Delta_p(d) \geq A_{a,r}h/2$ if also $h \leq A_{a,r}/(2b_{a,r})$. The least eigendirection is simple on the displayed neighbourhood; with consistent sign its distance from the split null vector is at most $6p(m-1)h$.

The general quadratic constants are deliberately conservative. They do not replace the substantially better seven-point remainder and radius already proved in v0.2. The advance is exact first-order control in every cardinality and every equality type, rather than a numerical improvement of that seven-point radius.

For $m \geq 9$, with mean cross-distance one and $h = \|d - s\|_\infty$ still fixed, the lower coefficient simplifies to

$$A_{a,r} = \begin{cases} p_m \frac{m+1}{m(m-1)}, & m \text{ odd}, \\ \frac{p_m}{m-2}, & m \text{ even}, a = r, \\ \frac{p_m}{m}, & m \text{ even}, r = a + 2. \end{cases}$$

Thus the two even equality types have different local condition constants. The metric normalisation and the norm used here are part of the statement; these constants are not invariant under changing that convention.

3. The exact tangent cone

At s , the only active triangle and Ptolemy inequalities are

$$u_{i;bc} = x_{ib} + x_{ic} - t_{bc} \geq 0,$$

$$v_{ij;bc} = x_{ib} + x_{ic} + x_{jb} + x_{jc} - 2y_{ij} - t_{bc} \geq 0,$$

for $i, j \in A$, $b, c \in B$, with unordered distinct pairs. There are $a \binom{r}{2}$ triangle rows and $\binom{a}{2} \binom{r}{2}$ Ptolemy rows. Denote their integer matrix by D . If w is the cross-edge indicator, direct counting gives

$$D^\top \mathbf{1} = a(r-1)w - r(r-1)\mathbf{1}_{AA} - \frac{a(a+1)}{2}\mathbf{1}_{BB}.$$

Hence $\ell(e) = \mathbf{1}^\top D e$ when $w^\top e = 0$.

The exact metric inequalities give $u \geq 0$ and

$$v_{ij;bc} \geq -3h^2,$$

because the omitted quadratic term is

$$x_{ib}x_{jc} + x_{ic}x_{jb} - y_{ij}t_{bc}.$$

All inactive inequalities have strictly positive slack at s .

Lemma 3.1 (every cone direction is geometrically attainable). Let $D e \geq 0$, $w^\top e = 0$, and $\|e\|_\infty \leq 1$. Let e^{in} equal -1 on within-part edges and zero on cross-edges. Then

$$d_t = s + t e + 4t^2 e^{\text{in}}, \quad 0 < t \leq 1/20,$$

is a positive Ptolemaic metric with mean cross-distance one.

Proof. On triangle rows $D e^{\text{in}} = 1$ and on Ptolemy rows $D e^{\text{in}} = 3$. Consequently the active triangle slack is at least $4t^2$, and the linear part of each active Ptolemy slack is at least $12t^2$. The absolute quadratic contribution is at most $3(t + 4t^2)^2 \leq (108/25)t^2 < 12t^2$. Every entry changes by at most $t + 4t^2 \leq 3/50$. An inactive triangle slack changes by at most $9/50 < 1$. An inactive Ptolemy slack, a signed sum of three products, changes by at most $3(4 \cdot 3/50 + (3/50)^2) < 1$. The original inactive slacks are at least one, so all remain positive. Positivity of distances and the normalisation are immediate. Conversely, the derivative of any differentiable feasible curve satisfies the linear inequalities. Thus this is the exact tangent cone, not merely a relaxation. $\square$

4. The sharp linear programme and its universal certificate

Theorem 4.1 (exact cone norm). For every binding pair in Section 2,

$$De \geq 0, \quad w^\top e = 0 \implies c_{a,r} \|e\|_\infty \leq \ell(e) \leq \frac{ar(r-1)(3a-1)}{4} \|e\|_\infty.$$

Both constants are attained by cone directions.

The upper bound follows from the explicit coefficients of ℓ , and e^{in} attains it. The lower bound has a finite rational certificate for every edge f and sign $\sigma \in \{-1, 1\}$:

$$\boxed{D^\top \eta + \mu w = D^\top \mathbf{1} - c_{a,r} \sigma \epsilon_f, \quad \eta \geq 0.} \quad (\text{C})$$

The supplied certificates additionally satisfy $\eta_j \leq 2$ on every Ptolemy row. Consequently the same identity for a nonlinear feasible metric gives

$$\boxed{\ell(e) \geq c_{a,r} h - 6 \binom{a}{2} \binom{r}{2} h^2.} \quad (4.1)$$

Indeed choose $\sigma e_f = h$. Triangle slacks are nonnegative, Ptolemy slacks are at least $-3h^2$, and the sum of their certificate weights is at most $2 \binom{a}{2} \binom{r}{2}$.

4.1 Exact description of the certificate families

The certificate is invariant under the stabiliser of the selected edge. For an anchor edge distinguish its two anchors; for a cross edge distinguish its anchor and pole; for a pole edge distinguish its two poles. A triangle row is classified by the number of distinguished anchors and poles it contains. A Ptolemy row is classified in the same way. Denote these row orbits by (u, i, j) and (v, i, j) .

The data in `certificates/symbolic_duals.json` give eighteen explicit rational-function vectors $\eta(k)$ and scalars $\mu(k)$: three families $(a, r) = (k, k + d)$, $d = 0, 1, 2$, three edge types and two signs. The domains are $k \geq 5$ for $d = 0$ and $k \geq 4$ for $d = 1, 2$. `CERTIFICATE_TABLES.md` prints all these expressions. No linear-programming solver is required to verify them.

Indispensable proof supplement. The complete tables, both JSON certificate collections, corrected verifiers, pinned dependencies and checksum manifest accompany this version in the archived research package [SUPP]. They are part of the proof of Theorem 4.1, not optional numerical illustrations. The eighteen symbolic certificates establish universal parameter ranges; sampled dimensions alone do not.

For clarity, one extremal certificate, the negative pole-edge certificate in the balanced family (k, k) , is

Row orbit	Weight
$(u, 0, 0)$	$k/(k-3)$
$(u, 0, 1)$	0
$(u, 0, 2)$	0
$(v, 0, 0)$	$(k^2 - 4k - 3)/[(k-3)(k-1)]$
$(v, 0, 1)$	$(k+1)/(k-1)$
$(v, 0, 2)$	$1/(k-1)$

Here $\mu = 1$ and $c = k^2/2$. At $k = 5 + x$, every weight is nonnegative for $x \geq 0$; for example $k^2 - 4k - 3 = x^2 + 6x + 2$. All Ptolemy weights are at most two.

For each of the eighteen vectors, the symbolic verifier constructs the orbit matrix directly by counting. If s_A, s_B distinguished vertices occur in the two parts, an orbit has multiplicity

$$\binom{s_A}{i} \binom{a-s_A}{n_A-i} \binom{s_B}{j} \binom{r-s_B}{2-j},$$

where $n_A = 1$ for a triangle row and 2 for a Ptolemy row. The coefficient of an edge orbit in $D^\top \eta$ is its total coefficient in a representative row, times this multiplicity, divided by the size of the edge orbit. Symmetry proves this counting formula. Substitution verifies (C) as a rational-function identity. Nonnegativity of η and of $2 - \eta$ for Ptolemy rows is checked by substituting $k = k_0 + x$: each numerator and denominator has nonnegative rational coefficients and the denominator has positive constant term. These are certificates on the entire half-line $x \geq 0$, not interpolation from sampled dimensions.

The six small pairs have thirty-six rational certificate vectors in `small_duals.json`, again covering the three edge types and both signs, all satisfying the same Ptolemy weight bound. A separate standard-library implementation enumerates the original labelled rows and verifies the identities without using symbolic orbit counting. It also evaluates the universal certificates at selected dimensions as a separate implementation check. The symbolic positivity certificates, not those finite evaluations, justify arbitrary cardinality.

4.2 Primal sharpness certificates

Distinguish two poles. Give every anchor the same cross perturbation u towards these two poles and $v = -2u/(r-2)$ towards the others. Give all anchor edges the same perturbation Y , the distinguished pole edge perturbation -1 , each pole edge with exactly one distinguished endpoint perturbation t_1 , and every other pole edge perturbation t_2 .

For the six small pairs use

$$u = -\frac{r-2}{3r-4}, \quad v = \frac{2}{3r-4}, \quad Y = -\frac{r-4}{2(3r-4)}, \quad t_1 = -\frac{r-4}{3r-4}, \quad t_2 = \frac{4}{3r-4}.$$

These entries have maximum absolute value one, obey $De \geq 0$ and the normalisation, and give

$$\ell(e) = \frac{ar\{(a-1)(r-2)(r-3) + 4\}}{4(3r-4)} = c_{a,r}.$$

There is no t_2 edge when $r = 3$; the unused value causes no issue.

For the balanced family $a = r = k \geq 5$ use

$$u = -\frac{r-2}{4(r-1)}, \quad v = Y = \frac{1}{2(r-1)}, \quad t_1 = -\frac{r-2}{2(r-1)}, \quad t_2 = \frac{1}{r-1}.$$

The six row-orbit slacks, in increasing numbers of distinguished poles, are $0, r/[4(r-1)], r/[2(r-1)]$ for triangles and $0, 0, 0$ for Ptolemy rows. Every entry has absolute value at most one, and direct substitution gives $\ell = ar/2 = c_{a,r}$. In the remaining two infinite families, simply shorten one pole edge by one and leave every other perturbation zero; this gives $\ell = a(a+1)/2 = c_{a,r}$. Thus the lower bound is attained. Lemma 3.1 integrates these directions into true metrics, proving that the linear-programming constants are the geometric first-order constants. $\square$

5. From the cone to the spectral gap

The exceptional constants reflect a change of the optimal face, not a different choice of normalisation. In the six small pairs, shortening the distinguished pole edge is accompanied by compensating cross-edge and anchor-edge changes. Their cost is smaller than that of shortening that edge alone. For example, at $(3, 5)$ this gives $60/11 < 6$. In the stable unbalanced families the single-edge direction is already optimal. In the balanced family the optimum continues to distribute the perturbation, with all Ptolemy rows tight in the displayed stable direction. Its dual certificate requires $k \geq 5$: the numerator $k^2 - 4k - 3$ is negative at $k = 4$. These explicit primal and dual facts explain why the small regimes require separate certificates; they are not a claim of uniqueness of the extremising direction.

The null eigenvalue at s is simple on $\mathbf{1}^\perp$, and the other eigenvalues are at least one. Put $H = -\Pi d^p \Pi$ on that space, and expand $H = H_0 + H_1 + H_2$. Because $0 < p < 2$, $h \leq 1/8$, and the reference distances are one or

two, entrywise Taylor expansion gives

$$\|H_1\| \leq 2p(m-1)h, \quad \|H_2\| \leq p(m-1)h^2, \quad \|H - H_0\| \leq 3p(m-1)h =: \varepsilon.$$

The second derivative bound used here is $|p(p-1)x^{p-2}|/2 \leq p$ on $x \geq 7/8$ when $0 < p < 2$ and x lies within $1/8$ of one or two. This loose bound also covers $p = 1$.

Direct expansion against the unit null vector gives

$$z_*^\top H_1 z_* = -\frac{2p}{m} \left[\frac{r}{a} \sum y_{ij} + \frac{aT}{2r} \sum t_{bc} \right] = \gamma \ell(e).$$

The cross term vanishes because the sum of cross perturbations is zero. This does not mean that individual cross edges are irrelevant: the active constraints couple them to within-part changes.

If $\varepsilon \leq 1/4$, the other eigenvalues remain at least $3/4$, and the least eigenvalue remains in $[-1/4, 1/4]$. Write the matrix in the decomposition $\mathbb{R}z_* \oplus z_*^\perp$. The scalar Schur equation gives

$$0 \leq z_*^\top H z_* - \lambda_{\min}(H) \leq \frac{\varepsilon^2}{1 - 2\varepsilon} \leq 2\varepsilon^2.$$

For the corresponding signed unit eigenvector, the block equation bounds the tangent of its angle from z_* by $\varepsilon/(1 - 2\varepsilon)$. The distance between the unit vectors is no larger than that tangent, hence at most 2ε . This is the elementary simple-eigenvalue estimate underlying standard perturbation theory [K]. Therefore

$$\gamma \ell(e) - [p(m-1) + 18p^2(m-1)^2]h^2 \leq \Delta_p(d) \leq \gamma \ell(e) + p(m-1)h^2.$$

Insert (4.1) and the upper cone bound. Since

$$6\gamma \binom{a}{2} \binom{r}{2} = \frac{3pr(a-1)}{m},$$

Theorem 2.1 follows. The extremal curves of Section 4 have $h(t) = t + O(t^2)$ and $\ell(e(t)) = c_{a,r}t + O(t^2)$ or its upper analogue, proving sharpness.

This proof is local and self-contained at the specified split metrics. It gives local negative type without assuming the global companion theorem. The companion and v0.2 remain premises only when one concludes that every possible global equality metric has one of these local models.

6. Sensitivity of the metric's own critical exponent

Let

$$\wp(d) = \sup\{q > 0 : d \text{ has } q\text{-negative type}\}.$$

This is the exponent of the particular metric, not the universal minimum over all metrics of its cardinality.

Relation to earlier exponent and gap results. Sánchez [S, Theorem 2.3 and Corollary 2.4] characterises the finite-metric endpoint using the distance matrix and its inverse on the constant vector. Thus detecting the critical exponent through loss of matrix strictness is established machinery. His complete-bipartite path metrics are not the complete split metrics here. Li–Weston [LW, Theorem 3.3] give quantitative extensions of the admissible exponent from a positive negative-type gap, as well as strict descent and endpoint non-strictness. The contribution here is the exact Ptolemaic tangent cone, its attained extremal directional constants, and their all-cardinality formulas, not a new general matrix characterisation.

Gap normalisations must also be distinguished. Our Δ_p minimises $-z^\top d^p z$ over zero-sum vectors with $\|z\|_2 = 1$. The normalised-simplex gap in [LW] uses positive and negative weights separately summing to one (equivalently $\|z\|_1 = 2$), with its associated quadratic-form factor. A rooted Gramian has yet another coordinate convention; Robertson [R, Sections 1–2] discusses its spectral relation to negative-type gaps for

ultrametric spaces. These quantities express related strictness properties but their numerical constants cannot be substituted without a normalisation conversion.

Theorem 6.1 (critical-exponent expansion). At every split covered by Theorem 2.1, for Ptolemaic $d \rightarrow s$ in mean-cross-distance normalisation,

$$\wp(d) - p = \frac{m}{r(a+1)\log 2} \Delta_p(d) + O(h^2) = \frac{2p\ell(e)}{ar(r-1)(a+1)\log 2} + O(h^2).$$

The constants hidden in $O(h^2)$ depend on the fixed split. Consequently the sharp lower and upper limits of $(\wp(d) - p)/h$ are

$$\frac{2pc_{a,r}}{ar(r-1)(a+1)\log 2}, \quad \frac{p(3a-1)}{2(a+1)\log 2}.$$

Proof. In positive distance coordinates, $-\Pi d^q \Pi$ is real analytic jointly in d, q . Its simple eigenvalue through zero at (s, p) has exponent derivative

$$\partial_q \lambda(s, p) = -\frac{a(r-1)T}{m} \log 2 = -\frac{r(a+1)}{m} \log 2 =: -\Lambda < 0.$$

The implicit-function theorem produces a unique nearby zero $q_c(d)$, with other restricted eigenvalues positive. The least eigenvalue changes from positive to negative as q increases through this root. Strict descent of negative type [LW] prevents another, larger negative-type interval: negative type at any larger exponent would imply strict negative type at an exponent immediately above the displayed root, a contradiction. Hence $\wp(d) = q_c(d)$. Expansion at $q = p$ gives

$$0 = \Delta_p(d) - \Lambda(q_c - p) + O(h|q_c - p| + (q_c - p)^2),$$

and $q_c - p = O(h)$ gives the first formula. Theorem 2.1 and its sharp curves give both limits. $\square$

For seven points these limits are

$$\frac{p_7}{24 \log 2} \quad \text{and} \quad \frac{p_7}{\log 2}.$$

This identifies how much extra snowflaking power a near-extremal metric actually permits. It is not an algorithmic complexity statement. The expansion concerns a fixed cardinality and normalisation; no uniform remainder over all m is claimed.

For large m , $p_m \sim 4/(m \log 2)$. The sharp lower gap coefficient is asymptotic to $4/(m^2 \log 2)$, while the upper coefficient tends to $3/(2 \log 2)$. Their ratio grows as $3m^2/8$. The sharp lower critical-exponent coefficient is asymptotic to $16/[m^3(\log 2)^2]$, whereas its upper coefficient is asymptotic to $6/[m(\log 2)^2]$. These follow by substitution, not by extrapolating numerical data.

7. Small-cardinality equality and the change of stability order

Proposition 7.1 (three and four points). At $P(3) = 2$, the equality metrics are precisely the three distinct collinear points, equivalently the positive triples whose longest distance is the sum of the other two. At $P(4) = \log_2 3$, the equality metrics are precisely the similarities of CS(1, 3) and CS(2, 2).

Proof. The three-point assertion is the Gram determinant criterion for a Euclidean triangle. For four points put $p = \log_2 3$. The negative-type endpoint is [B, Theorem 7.16]; every three-point metric has strict negative type at this $p < 2$ by [LW]. Thus an equality vector has full support.

If its sign pattern is 1 + 3, normalise the positive coefficient to one and write the negative magnitudes as $c_i > 0$, $\sum c_i = 1$. Let the three root lengths be ℓ_i . Ercan's scalar inequality [E, Lemma 2.1] with $2^p - 2 = 1$ gives

$$(\ell_i + \ell_j)^p \leq \ell_i^p + \ell_j^p + (\ell_i \ell_j)^{p/2},$$

with equality only when $\ell_i = \ell_j$. The gap is a sum of the triangle-power slacks, these scalar slacks and

$$\frac{1}{2} \sum_{i < j} (c_i \ell_i^{p/2} - c_j \ell_j^{p/2})^2.$$

All weights are positive. Equality therefore forces equal root lengths, leaf distances equal to their sums, and equal negative weights: the equal-arm star $\text{CS}(1, 3)$.

For a $2 + 2$ vector, negative type implies $d^P z = \lambda \mathbf{1}$. The potential λ is nonzero: otherwise Ercan's transport at any vertex would produce a nonzero equality vector supported on only three points of an inverted Ptolemaic metric, contradicting their strictness. Choose a vertex whose coefficient has the sign of λ . Its transported sign flips, giving a $1 + 3$ pattern after a global sign change. The inverted metric is the equal-arm star. Inverting back at the relevant independent vertex gives $\frac{1}{2} \text{CS}(2, 2)$, hence the second similarity type. Conversely both displayed metrics attain equality by direct substitution. The transport and the sign-selection mechanism are [E, Theorems 8.1–8.2]; only their equality specialisation is being used. $\square$

This elementary consequence does not settle five-point equality. In particular we do not replace Ercan's system of five-point equality conditions by an unsupported classification. The local theorem above is valid at $\text{CS}(2, 3)$, regardless of whether another five-point equality family exists.

Proposition 7.2 (a quadratic four-point direction). Put $p = \log_2 3$. Take the four-point star with root lengths $1 + t, 1 - t, 1$ and leaf distances their pairwise sums. Its mean root length is one and $h = |t|$. Then

$$\Delta_p(d_t) = \frac{p(2p - 3)}{8} t^2 + O(t^4).$$

The coefficient is positive because $3 > 2^{3/2}$. Therefore no positive local linear lower bound $\Delta_p(d) \geq ch$ can hold near the four-point star.

Proof. The equality vector is $(3, -1, -1, -1)/\sqrt{12}$; the other restricted eigenvalues are both three. The first scalar variation is zero. The direct quadratic Rayleigh term is $3p(p - 1)t^2/8$, and the second-order eigenvector correction is $p^2 t^2/8$. Their difference is the coefficient displayed. Swapping the first two leaves changes t to $-t$, so the simple eigenvalue is an even analytic function and the next term is $O(t^4)$. The symbolic verifier reconstructs these terms from the six raw distances, not from the asserted coefficient. $\square$

This is an obstruction to linear stability at four points, not a proof of a general quadratic lower stability theorem there. The different local order is a genuine distinction between that endpoint and the higher-cardinality split metrics.

8. Collision limits and an explicit smaller-cardinality gap

The classification can be stated cleanly on the compactification, without mislabelling collisions as equality metrics on distinct points.

Proposition 8.1 (kernel at a collision limit). Let a diameter-one Ptolemaic pseudometric on $m \geq 6$ labelled points have $N < m$ equivalence classes under zero distance. At exponent p_m , the null space of its restricted negative-type form consists exactly of vectors whose coefficient sum is zero separately in each equivalence class. Its dimension is $m - N$.

Proof. The quotient is an N -point Ptolemaic metric. Since $p_m < p_N$ for $N \geq 3$, the supplied threshold results and strict descent give strict p_m -negative type on the quotient. The $N = 2$ case is immediate. For any vector, the original quadratic form equals the quotient quadratic form applied to the class sums. On the zero-sum subspace it vanishes exactly when every class sum vanishes. $\square$

Thus, conditional on the global threshold and classification, the zero set on the diameter-one compactification comprises exactly the collision strata and the finitely many non-colliding split copies. The collision kernel has dimension $m - N$; a genuine extremiser has one nontrivial null direction, of full support.

A useful quantitative input for a future explicit global constant is available without a compactness argument.

Proposition 8.2 (explicit strict-descent gap). Suppose an N -point metric d , $N \geq 2$, has q -negative type and minimum distance at least $\eta > 0$. If $0 < p < q$ and $\alpha = p/q$, then

$$\Delta_p(d) \geq \frac{\eta^p}{2\Gamma(1-\alpha) [\log(2(N-1))]^\alpha}.$$

Proof. Realise d^q as squared Euclidean distances. The Gaussian kernel $K_t = (e^{-td_{ij}^q})$ is positive semidefinite: for a Euclidean realisation $d_{ij}^q = \|x_i - x_j\|^2$, expand $e^{2t\langle x_i, x_j \rangle}$ as a nonnegative sum of positive semidefinite Gram powers, then multiply by the positive diagonal factors $e^{-t\|x_i\|^2}$. For $t \geq t_0 = \log(2(N-1))/\eta^q$, every off-diagonal entry is at most $1/[2(N-1)]$, so $K_t \succeq I/2$ by diagonal dominance. For a unit zero-sum vector,

$$-z^\top d^p z = \frac{\alpha}{\Gamma(1-\alpha)} \int_0^\infty z^\top K_t z t^{-1-\alpha} dt \geq \frac{t_0^{-\alpha}}{2\Gamma(1-\alpha)}.$$

This is the asserted bound. The Gaussian positivity and integral representation are the standard strict-descent mechanism; the explicit tail estimate is the calculation used here. $\square$

For example, proper six-point submetrics at exponent p_7 admit this bound with $q = 1$, under the six-point threshold premise. This removes one source of non-effective constants in a quantitative inverse proof. It does not by itself yield a numerical C_η for the full seven-point problem. An explicit global constant and uniform collision-versus-rigidity estimates remain open tasks in this supplement.

9. Verification, novelty boundaries and next obligations

The supplied v0.2 archive replayed in full mode with the nested companion exact replay. Its exact, symbolic and LP reports matched the supplied files; the search trial records matched, while the search summary differed in runtime metadata. The unchanged replay receipt records the difference. We did not authenticate the reviewers or treat their reports as external specialist endorsements.

The new verification has three distinct roles:

1. The symbolic script verifies all eighteen universal rational-function certificates by direct orbit counts. Positive-coefficient expansions prove the required inequalities for every real parameter $k \geq k_0$, hence every integer cardinality in the stated families. It also re-derives the first-variation and four-point quadratic coefficients.
2. A separate standard-library rational script enumerates original labelled constraints, checks the thirty-six small certificates and seventy-two sampled evaluations of the general certificates, verifies all sharp primal directions and their exact Ptolemaic integrations, and checks the small-star examples. The sampled dimensions are implementation checks only.
3. Numerical perturbation and critical-exponent calculations corroborate the analytic formulas. The seeded convex-mixture trials combine only the displayed lower extremiser with the inward upper extremiser for each split; they do not sample the full feasible cone. They neither establish universal bounds nor substitute for an equality classification. High-precision records are retained separately.

In version 0.3.1 all proof-critical Python assertions have been replaced by explicit exception checks. Direct and replay entry points are tested under normal Python, $-O$, $-OO$, and both optimisation environment settings. Disposable negative controls alter a multiplier or the exact identity and regenerate the manifest, so checksum failure cannot mask the mathematical test. The retained mode-control report states the executed scope. These software checks do not formally verify the analytic proofs.

The dependency split is: Sections 3–5 use direct geometry, exact certificates and the written spectral estimate; Section 6 adds the classical endpoint/strict-descent results; Section 7 imports the public small-cardinality results and Ercan’s transport; Section 8 imports the companion’s threshold premises for the quotient

interpretation. Only the claim that the local models exhaust all global equality cases additionally uses the retained v0.2 classification.

The certificate tables are part of the proof evidence, not merely exploratory solver output. No optimisation solver is needed by either exact verifier. Numerical linear programming was used during discovery, and its exploratory scripts and logs are retained under *provenance/discovery/* rather than described as proof certificates.

The following remain unproved here: a classification of five-point equality, a full quadratic stability theorem at four points, a numerical global separated-class constant, or a dimension-uniform sharp second-order remainder. The critical-exponent theorem has a proved fixed-cardinality expansion, not a numerical claim about all neighbourhood sizes. The local all-size result does not require the unrefereed companion; the global equality interpretation does. Historical priority has not been established for any new lemma.

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